F78AA1 ACTUARIAL & FINANCIAL MATHEMATICS A

DECEMBER 2015

SOLUTIONS

Q1.(a) Let $t=0$ be 1/1/15, and choose time unit one year.

$+100\, v(0,0.5) = 100(1.04)^{-0.5} = £98.06$

$+100\, v(0,1.5) = 100(1.04)^{-1}(1.02)^{-1} = £94.27$

$-60\, v(0,2.75) = -60(1.04)^{-1}(1.02)^{-1.5} = -£53.83$

$-60\, v(0,3.5) = -60(1.04)^{-1}(1.02)^{-4}(1.01)^{-2} = -£52.25$

$-60\, v(0,4.5) = -60(1.04)^{-1}(1.02)^{-4}(1.01)^{-6} = -£50.21$

Total £36.04

1 mark for correct times; 3 marks for correct discount factors; 1 mark for correct answer.

[5 marks]

(i) P.V. in (a) $\times\, v(5.5,0)$

$= 34.04(1.04)(1.02)^{4}(1.01)^{10} =$ £44.82

[2 marks]

[Total 7 marks]

Q2 (a) $\ddot{s}^{(i)}_{\overline{n}|}$ is the accumulation at the end of n time units, at effective interest of i per time unit, of an annuity of 1 per time unit, payable at the start of the first, second, ... n^{th} time units

[1 mark].

(Credit will be given for alternatives that are equally complete and precise.)

(b) $\ddot{s}^{(i)}_{\overline{n}|} = (1+i)^n + (1+i)^{n-1} + \ldots + (1+i)$ [1]

or $(1+i)\ddot{s}^{(i)}_{\overline{n}|} = (1+i)^{n+1} + (1+i)^n + \ldots + (1+i)^2$ [1]

or $(1+i)\ddot{s}^{(i)}_{\overline{n}|} - \ddot{s}^{(i)}_{\overline{n}|} = (1+i)^{n+1} - (1+i)$

ie $i\,\ddot{s}^{(i)}_{\overline{n}|} = (1+i)\left[(1+i)^n - 1\right]$

so $\ddot{s}^{(i)}_{\overline{n}|} = \frac{(1+i)}{i}\left[(1+i)^n - 1\right]$ [1]

$= \frac{(1+i)^n - 1}{d}$ since $d = \frac{i}{1+i}$ [1]

[4 mark]

[Total 5 mark]

Q3. The p.v. at $t=10$ year of the last five payments is

$$10{,}000\,(1.02)^{10}\left[\frac{1.02^0}{1.06}+\frac{1.02^1}{1.06^2}+\cdots+\frac{1.02^4}{1.06^5}\right]$$

↑ amount of 11th payment

$$=10{,}000\,(1.02)^9\left[\frac{1.02}{1.06}+\frac{1.02^2}{1.06^2}+\cdots+\frac{1.02^5}{1.06^5}\right]$$

$$=10{,}000\,(1.02)^9\;a^{j_1}_{\overline{5}|} \quad \text{where } 1+j_1=\frac{1.06}{1.02}=1.039216$$

$$=10{,}000\times 1.195093\times 4.4616$$

$$=53{,}320.27.$$

Hence the p.v. at $t=0$ of all the payments is

$$10{,}000\left[\frac{1.02^0}{1.04}+\frac{1.02^1}{1.04^2}+\cdots+\frac{1.02^9}{1.04^{10}}\right]+1.04^{-10}\times 53{,}320.27$$

↑ 36,021.26

$$=\frac{10{,}000}{1.02}\left[\frac{1.02}{1.04}+\frac{1.02^2}{1.04^2}+\cdots+\frac{1.02^{10}}{1.04^{10}}\right]+36{,}021.26$$

$$=\frac{10{,}000}{1.02}\;a^{j_2}_{\overline{10}|} \quad \text{where } 1+j_2=\frac{1.04}{1.02}=1.019608$$

$$=124{,}266.71.$$

Therefore the accumulation at $t=15$ is

$$124{,}266.71\,(1.04)^{10}(1.06^5)$$

$$=246{,}160.02$$

Q4 (a) (i) Repayment $X = \frac{200,000}{a_{\overline{30}|}^{0.07}} = £16,117.28$ [2]

(ii) Cap o/s $= X\, a_{\overline{20}|}^{0.07} = £170,746.69$ [2]

(i) Int. cont. $=$ Cap o/s $\times 0.07 = £11,952.27$ [1]

[5 marks]

(b) After three years, cap o/s will be

$170,746.69\,(1.07)^3 - 25,000\, s_{\overline{3}|}^{0.07}$ [2]

$= £128,799.53$

Effective interest is 0.005634 per month [1]

New repayment is:

$\frac{128,799.53}{a_{\overline{204}|}^{0.005634}}$ [1]

$= £1,065.58$ [1]

[5 marks]

[Total 10 marks]

Q5. (a) Test for CGT : $\frac{0.04 \times 0.8}{1.02} = 0.031373 < 0.034408$ [1]

↑ effective interest per half-year [1]

so CGT will be payable. Hence

$$P = 4 \times 0.8 \times a_{\overline{50}|}^{0.034408} + (102 - 0.2(102 - P))(1.07)^{-25}$$ [2]

↑ 23.70816 ↑ 0.184249

$$\Rightarrow P(1 - 0.036850) = 75.8661 + 15.0347$$ [2]

$$\Rightarrow P = £94.38$$ [1]

[6 marks]

(b) If a bond is bought at 100, pays net coupons of g per time unit, and is sold or redeemed at 100, then the yield obtained is g per time unit [1]

So if a bond is bought and sold at $P \neq 100$, and pays net coupons of g per time unit, it is equivalent to buying and selling at 100 a bond paying coupons of $\frac{100}{P} \times g$ per time unit [1]

Here, net coupon = 3.2 per half year, $P = 94.38$,

so yield is $\frac{100}{94.38} \times 3.2 = 0.033905$ per half-year, [1]

or 0.069 p.a.

[3 marks]

[Total 9 marks]

Q6(a) Equation of Value:

$$f(i) = 100(1+i)^4 + 20(1+i)^3 - 50(1+i) - 90 = 0$$ [2]

Trial and error:

$f(0.04) = -2.517$

$f(0.05) = +2.203$ [2].

Linear interpolation

$$i^* \cong 0.04 + 0.01 \frac{2.517}{2.517 + 2.203}$$

$$= 0.0445$$ [1]

[5 marks]

(b) $$(1+TWRR)^4 = \left(\frac{102}{100} \times \frac{133}{122} \times \frac{129}{133} \times \frac{90}{79}\right)$$ [1]

$$\Rightarrow TWRR = 0.052838$$ [1]

[2 mark]

[Total 7 mark]

Q7. Method A

Gross coupon is 0.4m p.a.

Investor must borrow 0.4m at $t=1$ to pay first coupon. [1]

At time $t=2$, repay this loan so net profit is

$$1.5 - 0.4 - 0.4(1.07) = 0.672 \text{ m}.$$ [1]

At time $t=3, \ldots, 8$, net profit is $1.5 - 0.4 = 1.1$m [1]

At time $t=8$, repay $5(1.02) = 5.25$ m. [1]

$\Rightarrow$ Accumulated profit is $0.672(1.06)^6 + 1.1\, s_{\overline{8}|}^{0.06} - 5.25$

$= 3.376$ m. [1]

[5 marks].

Method B.

Level repayment is $X = \dfrac{5}{a_{\overline{8}|}^{0.07}} = 0.837339$ m. [1].

Investor must borrow 0.837339 m. at $t=1$ to make repayment

At time 2, repay this loan so net profit is

$$1.5 - 0.837339 - 0.837339(1.07) = -0.233292 \text{ m}$$ [1]

Hence must borrow 0.233292 at $t=2$.

At $t=3$ repay this loan so net profit is

$$1.5 - 0.837339 - 0.233292(1.07) = 0.413039 \text{ m}$$ [1]

At $t=4, 5, 6, 7, 8$, net profit is $1.5 - 0.837339 = 0.662661$ m [1]

So accumulated profit is

$$0.413039(1.06)^5 + 0.662661\, s_{\overline{5}|}^{0.02}$$

$= 4.288221$ m. [1]

[5 marks].

Method C

At $t-1$, cap o/s has increased to $5(1.07) = 5.35$ m. [1]

Loan will be repaid at time k where

$$1.5 s_{\overline{k}|}^{0.07} \geq 5.35 (1.07)^{k-1} \quad [1]$$

$$\Rightarrow \quad \frac{(1.07)^k - 1}{0.07} \geq 3.5667 (1.07)^{k-1}$$

$$\Rightarrow \quad (1 - 0.24967)(1.07)^{k-1} \geq 1$$

$$\Rightarrow \quad (1.07)^{k-1} \geq 1.33274$$

$$\Rightarrow \quad k-1 \geq \frac{\log(1.33274)}{\log(1.07)} \quad [2].$$

$$\Rightarrow \quad k-1 \geq 4.24$$

Hence DPP $= 1 + 5 = 6$ years.

Net profit at $t = 6$ is

$$1.5 s_{\overline{5}|}^{0.07} - 5.35 (1.07)^5 = 1.1224 \text{ m}. \quad [1]$$

Hence accumulated profit is

$$1.1224 (1.06)^2 + 1.5 s_{\overline{2}|}^{0.06} = 4.3511 \text{ m} \quad [1]$$

Hence, prefer method C.

[6 marks]

[Total 16 marks]